

Learning to Make the Working Catalyst: Autonomous Synthesis–Activation Co-Design

Intellectual Merit

A catalyst is not necessarily the material that enters the reactor. Song et al. linked inherited precursor architecture to n-propanol selectivity under one activation protocol [1]. In a different reactor, Li et al. obtained Cu adparticles under CO reduction, but grain-boundary-rich surfaces under nitrogen that later restructured during CO operation [2]. Does activation overwrite precursor structure, or can both histories shape the working catalyst? Building on robotic gas-fed testing [3], I will develop a self-driving laboratory that learns these relationships while choosing what to make, measure, or repeat. CO reduction is the model reaction; transfer to CO₂ reduction is not assumed.

Hypotheses. Activation-created features do not uniformly erase precursor architecture: I predict gas-dependent retention of structural and product contrasts. Separately, adaptive characterization and replication will reach a specified precision on the synthesis–activation interaction curve at lower resource cost, without worse validation error or calibration. The two hypotheses will be tested independently.

Aim 1: Determine what activation allows a precursor to retain.

I will robotically spray copper nitrate on carbon paper and apply Joule heating [1], recording calibrated temperature traces. A pilot will cross two dwell times, at fixed peak temperature and cooling, with CO or N₂ activation. Total flow and the current–time program will be shared; all electrodes then enter identical CO-fed membrane-electrode-assembly (MEA) tests. Structural checks will assess comparable reduction endpoints; equal charge alone will not establish equal conversion. Independent coating/heating runs will be replicated, with day and channel balanced. Pilot variance and a predeclared meaningful interaction will determine confirmatory replication.

Diffraction and microscopy before activation, at its endpoint, and after operation will track oxide fractions, coherent-domain dimensions and particle connectivity. Blinded transmission electron microscopy (TEM) will assess selected surface and grain/subgrain structures; diffraction width alone cannot identify boundaries. Air-free transfer will be checked against deliberate exposure [4]. Destructive or perturbing measurements use identically prepared companions, with mismatch estimated experimentally. Heated-support controls, loading, ICP-OES contamination checks and wetting/CO-access tests constrain alternative pathways [5]. A cooling-history comparison, accepted only with matched oxide fractions, will challenge the architectural interpretation. Transport can mediate a structural effect rather than invalidate it.

The primary outcome is late-window n-propanol partial current density at fixed total current in a two-hour screen; an early window tests evolution. Calibrated gas chromatography, liquid collection and quantitative nuclear magnetic resonance will check volatile-product recovery, collection delay, outlet flow and crossover [1,8]. Activation products are collected separately. Product ratios remain diagnostic; absolute production, charge balance and voltage remain reported. Selected 50-hour tests assess persistence, not industrial lifetime. The decisive comparison tests whether the precursor contrast changes with activation, then validates a predicted structure–performance response in a new preparation. A processing interaction alone cannot establish a specific active-site mechanism.

Interpretation. Gas-dependent retention with a replicated product interaction supports co-design. Equivalent structural and product contrasts across activation gases argue against that interaction within predeclared meaningful bounds. Retained structure with convergent products limits its relevance to the measured output; imprecise results remain unresolved.

Aim 2: Build a laboratory that chooses what to make, measure, or repeat.

After assay validation, I will expand dwell time and activation-gas composition within the tested window. Let $\vartheta(t)$ be the CO-versus-N₂ effect on late-window output at dwell t , minus that effect at a reference dwell. This interaction curve is the common target for decisions and scoring. Adaptive characterization and replication exist [6,7]; here the challenge is delayed, destructive assays on imperfectly matched companions. A hierarchical joint model will link processing, structure and performance, including preparation/reactor variation and measurement error. Pretest companion data can inform predictions without proving mediation; post-test data support interpretation.

The controller selects a new preparation–activation pair, independent repeat or validated structural assay by expected reduction in integrated posterior variance of $\vartheta(t)$ per resource cost, using a fixed dwell grid and weights. Cost includes specimen preparation, instrument occupancy, analysis and delay; expensive assays must earn their use. A scheduler executes robotic coating/heating, cassette handling, gas switching and testing; delayed results update subsequent batches. Calibrations and safety checks are mandatory. Equipment faults invalidate measurements; reproducible material failures remain feasibility outcomes.

I will collect a separate, policy-independent replicated grid with paired structural/product data, failures and costs. Replay reveals only acquired measurements; independent validation preparations remain hidden. A space-filling design and an adaptive synthesis–activation search with fixed characterization/replication share models, candidates, starting data, controls and budgets. The primary score is cost to the predeclared integrated-variance threshold for $\vartheta(t)$, subject to validation error and calibration limits on that same curve. Prospective campaigns interleaved across days test physical performance beyond replay. Condition and preparation holdouts distinguish transfer from reproducibility. Removing structural inputs tests their value; unresolved comparisons are not wins.

Preparation and feasibility. My Yushin-group nanomaterials synthesis and robotic coating, electrochemistry and ICP-OES work at Lila Sciences prepare me for this project. Year 1 validates precursor reproducibility, recovery, throughput and interfaces; Year 2 closes the physical loop and builds the reference library; Year 3 tests new batches and persistence. I will seek a host with existing gas-fed testing and shared microscopy/NMR, extending one station before parallelizing. Unattended CO operation requires professionally reviewed interlocks and commissioning. Operando beamtime is optional.

Broader Impacts

This project will support electrified chemical synthesis and reusable autonomous-research infrastructure. Subject to host agreements, I will release nonproprietary interfaces, fixtures, calibrations and datasets preserving failures and sample histories. I will develop “Build the Experiment, Not Just the Model,” a laptop-based undergraduate module that replays the campaign without robots or hazardous gases. Students will choose whether to make, measure or repeat under a resource budget. I will seek a course partner, pilot with approximately twenty students, and assess reasoning with a pre/post rubric on confounders, uncertainty and decision quality. An undergraduate mentee will help curate accessible documentation and exercises, linking energy-materials education to autonomous science.

References

[1] J.-Y. Song et al., *Adv. Mater.* 37, e08900 (2025). [2] J. Li et al., *Nat. Commun.* 9, 4614 (2018). [3] A. Soni et al., *Matter* 9, 102519 (2026). [4] Y. Yang et al., *Nature* 614, 262–269 (2023). [5] Q. Xu et al., *Nat. Catal.* 6, 1042–1051 (2023). [6] S. Ament et al., *Sci. Adv.* 7, eabg4930 (2021). [7] M. Binois et al., *Technometrics* 61, 7–23 (2019). [8] E. L. Clark et al., *ACS Catal.* 8, 6560–6570 (2018).